

DevOps Intern Assignment – 10xConstruction

1. Context

You are working with an edge computing robot module with:

- 2-core CPU at approximately 2 GHz
- 500 MB usable RAM
- Docker installed

Your task is to deploy and optimize a resource-constrained observability stack on the device. A small Python sensor service is provided in the accompanying ZIP. It contains intentional inefficiencies that cause memory usage, CPU spikes, and scraping failures. You must containerize it, optimize it, and make it observable.

2. Objective

Build a functional observability setup including:

- The provided Python sensor service
- A Prometheus-compatible metrics collector
- A visualization solution (Grafana or lighter alternative)

All components combined must consume less than 300 MB RAM.

You must debug and fix intermittent scrape failures caused by performance issues in the service.

3. Required Tasks

3.1 Deploy the Provided Sensor Service

- Containerize and run the provided [Python service](#)
- Identify CPU spikes, memory spikes, and scrape delays
- Fix or reduce root-cause inefficiencies

3.2 Metrics System (Prometheus or Alternative)

Choose one:

Option A: Prometheus with memory optimizations:

- Short retention
- WAL compression
- Reduced scrape frequency

Option B: Prometheus-compatible lightweight alternative:

- VictoriaMetrics single node
- Mimir agent
- OpenTelemetry Collector with remote_write

You must justify your choice.

3.3 Visualization Layer

Choose one:

- Grafana
- Lighter UI (VictoriaMetrics)

- Static HTML dashboard
- Terminal dashboard

If choosing Grafana, justify how you stay below memory limits.

3.4 Custom Metric Requirement

Add one meaningful custom metric to the Python service, such as:

- Histogram of CPU spike duration
- Gauge of queue depth
- Counter for failed events

3.5 Final Optimized Docker Compose

Produce a final optimized Docker Compose setup that stays within the memory limit.

4. Performance Budget Report (1–2 pages)

Include:

- Memory usage of each component (before vs. after)
- Identified bottlenecks in the Python service
- Observability design decisions
- Justification of Prometheus and visualization choices
- The custom metric and reasoning
- One improvement if given one more week

5. Design Twist (Debugging Required)

The provided Python service MIGHT occasionally:

- Burn high CPU
- Allocate excess memory
- Cause Prometheus scrape timeouts and delays
- Produce inconsistent response times
- Cause dashboards to fail intermittently

You must solve these through:

- Code-level optimizations
- Metrics tuning
- Tool selection
- Memory budgeting
- Scrape interval optimization

Find the required [Files here](#) - DevOPS- Assignment - 10x Intern

Note: Use of AI tools to complete the assignment is discouraged, vibe coding of the entire assignment is frowned upon, intelligent use of AI tools to get ideas and solve the same is encouraged.

Submission : Make sure it is a private github repository , add tanay@10xconstruction.ai , piyush@10xconstruction.ai , tushar@10xconstruction.ai as collaborators. Attach a small video walkthrough of your assignment explaining the same , and your approach in the github readme or as a drive lin